Unit: 6 Unit 9: Data Statistical Analysis　　**Name:** ____________
Topic: Stem and Leaf Plots　　**Date:** __________

Advanced (Carolina Reaper)

Q1. What is the primary purpose of a stem-and-leaf plot?
(A) To display the mean of a dataset
(B) To show the distribution of a dataset
(C) To calculate the median of a dataset
(D) To create a box plot
(E) To find the mode of a dataset

Q2. Which of the following is a benefit of using a stem-and-leaf plot?
(A) It can only be used with small datasets
(B) It is difficult to compare multiple datasets
(C) It can help identify outliers in a dataset
(D) It is only used for categorical data
(E) It is not useful for skewness

Q3. What does the 'stem' represent in a stem-and-leaf plot?
(A) The first digit of each data point
(B) The last digit of each data point
(C) The middle digit of each data point
(D) The average of each data point
(E) The range of each data point

Q4. Which of the following datasets would be most suitable for a stem-and-leaf plot?
(A) A dataset with 5 values: $1, 2, 3, 4, 5$
(B) A dataset with 10 values: $10, 20, 30, 40, 50, 60, 70, 80, 90, 100$
(C) A dataset with 15 values: $1.1, 1.2, 1.3, 1.4, 1.5, 2.1, 2.2, 2.3, 2.4, 2.5, 3.1, 3.2, 3.3, 3.4, 3.5$
(D) A dataset with 20 values: $1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, 10000, 11000, 12000, 13000, 14000, 15000, 16000, 17000,$
(E) A dataset with 25 values: $0.1, 0.2, 0.3, 0.4, 0.5, 1.1, 1.2, 1.3, 1.4, 1.5, 2.1, 2.2, 2.3, 2.4, 2.5, 3.1, 3.2, 3.3, 3.4, 3.5, 4.1, 4.2, 4.3, 4.4, 4.5$

Q5. What is the 'leaf' in a stem-and-leaf plot?
(A) The last digit of each data point
(B) The first digit of each data point
(C) The average of each data point
(D) The range of each data point
(E) The median of each data point

Q6. Which of the following stem-and-leaf plots represents a skewed distribution?
(A) Stem: 1, Leaves: $2, 3, 4, 5, 6$
(B) Stem: 1, Leaves: $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$
(C) Stem: 2, Leaves: $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$
(D) Stem: 1, Leaves: $6, 7, 8, 9, 10$
(E) Stem: 1, Leaves: $0, 5, 10, 15, 20$

Q7. What does a stem-and-leaf plot with multiple stems indicate?
(A) A small dataset
(B) A large dataset
(C) A dataset with multiple modes
(D) A dataset with multiple medians
(E) A dataset with a large range

Q8. Which of the following is NOT a common use of stem-and-leaf plots?
(A) To compare the distribution of two datasets
(B) To identify the median of a dataset
(C) To calculate the mean of a dataset
(D) To display the distribution of a dataset
(E) To create a scatter plot

Open Ended Questions (FRQ)

Q9. Construct a stem-and-leaf plot for the dataset: $23, 11, 19, 17, 22, 26, 18, 24, 20, 16, 25, 21$

Q10. A stem-and-leaf plot is given below.

Chef's Tips
- Remind students to read each question carefully
- Emphasize the importance of labeling each part of the stem-and-leaf plot
- Encourage students to use the space provided to show their work